



Learner Guide

Navigating Digital Threats: Proactive Measures Against Cyber Frauds and Scams

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Version	Release date	Summary	Author
8.0	2024	Original reference material	Tertiary Infotech Academy
9.0	7 September 2026	Rebuilt from the v8 reference deck; added 2025 and first-half 2026 Singapore evidence, eight self-contained activities, current ScamShield guidance, layered controls and incident escalation practice.	Dr Alfred Ang

How to Use This Guide

This guide supports an eight-hour instructor-led WSQ course plus a one-hour assessment. It explains how scam approaches work, how to verify claims outside the attacker-controlled channel, how to protect accounts and payments, and how to preserve evidence and escalate an incident. Detailed activity procedures are collected here and repeated in each activity folder.

Learning Outcomes

- LO1 Identify potential cyber security risks and threats, including internet frauds and scams.
- LO2 Protect personal and business information from internet frauds and scams.
- LO3 Interpret and escalate cyber frauds and scams to relevant authorities.

Singapore Scam Evidence

SPF reported 37,308 scam cases and about S\$913.1 million in losses for 2025. For January to June 2026, scam cases fell to 16,821 and losses to about S\$410.6 million, but online platforms remained the first approach in 89.0% of scam cases. The operational lesson is to design controls around the first contact, the independent verification break, the payment path and the response timeline.

Primary references: [SPF Annual Brief 2025](#) | [SPF Mid-Year Brief 2026](#) | [ScamShield](#)

Topic 1 Identifying Cyber Fraud and Scams

A scam is best analysed as a chain: approach, pretext, pressure, requested action, transfer or access route, and concealment. Observable evidence matters more than intuition. The same pretext can move across email, calls, messaging, social media, QR codes and fake sites.

Pattern	Mechanism	Asset targeted	Verification break
Phishing	Spoofed email or site	credential or card capture	Use an official route obtained independently; inspect sender-domain mismatch; login link; authentication-code request
Smishing	Malicious SMS or chat	mobile-first credential capture	Use an official route obtained independently; inspect unknown sender; shortened link; delivery or refund pretext
Vishing	Impersonation call	coerced disclosure or transfer	Use an official route obtained independently; inspect caller-ID trust; authority claim; secrecy; stay-on-line pressure
Government impersonation	False official investigation	transfer to a safe account	Use an official route obtained independently; inspect threat of arrest; transfer demand; remote interview
Bank impersonation	False fraud alert	OTP or digital-token takeover	Use an official route obtained independently; inspect urgent reversal; app install; authentication request
Investment scam	Fabricated returns platform	repeated deposits and blocked withdrawal	Use an official route obtained independently; inspect guaranteed return; social proof; withdrawal fee
Job scam	Fake task or recruitment offer	advance payment and escalating deposits	Use an official route obtained independently; inspect easy income; group chat; pay to unlock commission
E-commerce scam	Non-existent or misrepresented goods	irreversible prepayment	Use an official route obtained independently; inspect off-

			platform chat; deposit; price anomaly
Tech-support scam	False device infection	remote access and payment	Use an official route obtained independently; inspect unsolicited alert; remote tool; refund or clean-up fee
Business email compromise	Compromised or spoofed mailbox	fraudulent supplier payment	Use an official route obtained independently; inspect changed beneficiary; urgency; process bypass
Romance scam	Engineered emotional trust	progressive financial extraction	Use an official route obtained independently; inspect rapid intimacy; crisis; avoidance of live verification
Malware-enabled scam	Sideloaded or disguised application	session theft and device control	Use an official route obtained independently; inspect unknown-source app; accessibility permissions; overlay
QR phishing	Malicious QR destination	hidden redirect to fake service	Use an official route obtained independently; inspect unverifiable code; payment/login request; domain mismatch
Account takeover	Stolen credentials or session	impersonation and onward fraud	Use an official route obtained independently; inspect new device; recovery change; unusual messages
Money-mule recruitment	Use of personal accounts to move funds	laundering and legal exposure	Use an official route obtained independently; inspect keep commission; receive and forward; secrecy
Lottery and prize scam	False reward notification	fee and identity extraction	Use an official route obtained independently; inspect unexpected win; tax or delivery fee; urgency
Loan scam	False low-cost credit offer	advance fee and identity theft	Use an official route obtained independently; inspect guaranteed approval; processing fee; unofficial channel
Deepfake impersonation	Synthetic voice or video	high-trust payment manipulation	Use an official route obtained independently; inspect unusual request; channel shift; no independent callback

Topic 2 Assessing and Protecting Information

Protection is layered. Identity assurance, device integrity, account security and payment controls should each interrupt a different part of the attack chain. A control is useful only when the learner can name the attack path it blocks, the evidence that shows it is enabled and the residual risk.

Independent verification

Move to contact data obtained outside the suspicious message. Breaks the scammer-controlled trust path. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Passkeys and phishing-resistant MFA

Bind authentication to the legitimate origin and device. Reduces credential replay and OTP interception. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Password manager

Generate unique credentials and match saved origins. Prevents password reuse and exposes look-alike domains. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Device patching

Apply supported operating-system and browser fixes. Removes known exploit paths used by malicious sites and apps. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Unknown-source restriction

Block sideloading and risky accessibility permissions. Interrupts malware-enabled scam installation. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Session review

Inspect devices, sessions, forwarding and recovery changes. Detects persistence after credential compromise. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Transaction alerts

Notify immediately on transfers, payees and card activity. Shortens detection and bank-contact time. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Transfer limits

Cap routine exposure and require deliberate changes. Reduces maximum loss from one compromised session. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Money Lock

Separate protected funds from digitally transferable balances. Limits funds reachable through online compromise. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Maker-checker approval

Separate request, verification and payment release. Stops a single compromised mailbox from authorising payment. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Beneficiary callback

Verify account changes through a known channel. Targets business email compromise and invoice diversion. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

ScamShield

Check and report suspicious numbers, messages and links. Adds community and SPF-informed detection signals. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Privacy minimisation

Limit public identity, role and relationship data. Reduces material available for tailored pretexts. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Offline backup

Keep tested, disconnected recovery copies. Reduces ransomware leverage and preserves business continuity. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Incident rehearsal

Practice evidence, bank, police and internal escalation. Reduces delay and omissions under pressure. Record configuration evidence without exposing passwords, OTPs, recovery codes or sensitive account numbers.

Topic 3 Responding and Escalating

Response priorities are time-sensitive: stop further engagement, protect funds, isolate suspected compromise, preserve evidence, secure accounts, assess the scope and notify the parties who can act. Keep a chronology and distinguish confirmed facts from assumptions.

Stop interaction

End calls and messages; do not follow new instructions. This prevents further coercion and data disclosure. Record the time, owner, evidence reference and outcome.

Protect money

Contact the bank through an official channel and request urgent action. This maximises the chance of stopping or tracing transfers. Record the time, owner, evidence reference and outcome.

Isolate compromise

Disconnect a suspected infected device while preserving state. This limits attacker access without erasing evidence. Record the time, owner, evidence reference and outcome.

Preserve evidence

Save originals, screenshots, headers, numbers, urls and transaction references. This supports bank, platform and police action. Record the time, owner, evidence reference and outcome.

Secure accounts

Change exposed credentials from a trusted device and revoke sessions. This closes reused credentials and stolen sessions. Record the time, owner, evidence reference and outcome.

Assess scope

Identify people, accounts, devices, data and payments affected. This sets severity and notification priorities. Record the time, owner, evidence reference and outcome.

Report platform abuse

Use the platform reporting path for the account, advert, message or listing. This enables removal and ecosystem disruption. Record the time, owner, evidence reference and outcome.

Use ScamShield

Check or report suspicious contact through official scamshield resources. This contributes signals and receives current advice. Record the time, owner, evidence reference and outcome.

Make a Police report

Provide a factual chronology and supporting evidence. This creates an official record and supports investigation. Record the time, owner, evidence reference and outcome.

Notify the workplace

Engage security, finance, legal and data-protection roles as applicable. This coordinates containment and statutory decisions. Record the time, owner, evidence reference and outcome.

Track recovery

Maintain actions, owners, reference numbers and outcomes. This prevents gaps and supports handoffs. Record the time, owner, evidence reference and outcome.

Review controls

Convert the incident into owned corrective actions. This reduces recurrence and tests whether controls work. Record the time, owner, evidence reference and outcome.

Detailed Activity Guides

Activity 1 Scam Signal Triage

Alignment: LO1 / A1 | Objective: Classify a suspicious contact using channel, claim, pressure, requested action and payment route.

Procedure

1. Open the supplied message pack and assign each item an evidence ID.
2. Record the claimed sender, channel, callback details, URL and requested action.
3. Mark authority, urgency, scarcity, fear, reward and secrecy pressure cues.
4. Identify the asset at risk: credentials, money, device access, identity or business data.
5. Trace the proposed payment or access route and note reversibility.
6. Score likelihood and impact from 1 to 5 using the worksheet rubric.
7. Choose block, verify, monitor or escalate and state the evidence for the choice.
8. Peer-review one classification and revise unsupported assumptions.

Evidence and acceptance

Completed signal matrix with risk score and decision rationale.

Acceptance: Every decision cites at least two observable indicators; no decision relies only on spelling or grammar.

Activity 2 Phishing URL and Header Analysis

Alignment: LO1 / A1 | Objective: Inspect a simulated phishing message without opening its link or attachment.

Procedure

1. Work only from the supplied offline email sample and URL text file.
2. Compare the display name with the From and Reply-To domains.
3. Read Received hops from bottom to top and identify the first untrusted handoff.
4. Check SPF, DKIM and DMARC results recorded in Authentication-Results.
5. Decompose the URL into scheme, host, registrable domain, path and query.
6. Flag punycode, look-alike characters, subdomain deception and redirect parameters.
7. Write an independent verification route using a known bookmark or official directory.
8. Capture the findings in the evidence template and assign a disposition.

Evidence and acceptance

Annotated header and URL decomposition with disposition.

Acceptance: The registrable domain and authentication result are stated correctly and no live suspect link is opened.

Activity 3 Independent Verification Drill

Alignment: LO1 / A1 | Objective: Break the scammer-controlled channel and verify a claim through trusted contact data.

Procedure

1. Select one impersonation scenario from the activity pack.
2. List every contact path supplied by the sender and treat it as untrusted.
3. Locate an official contact route from a saved statement, app, card or typed official domain.
4. Draft a neutral verification script that reveals no OTP, password or account PIN.
5. Verify the claimed event, transaction or officer identity through the official route.
6. Record who was contacted, when, which reference was checked and the response.
7. Decide whether to block, continue cautiously or escalate.
8. Explain which trust boundary changed when you moved to the independent channel.

Evidence and acceptance

Verification log and trust-boundary explanation.

Acceptance: No sender-provided telephone number, link or QR code is used for verification.

Activity 4 Account and Device Hardening

Alignment: LO2 / A2 | Objective: Apply layered safeguards that reduce credential, session and malware-enabled fraud.

Procedure

1. Inventory two high-value accounts and the recovery channels attached to them.
2. Replace any reused password with unique password-manager-generated credentials.
3. Enable a phishing-resistant sign-in method or app-based MFA where available.
4. Review active sessions, trusted devices, forwarding rules and recovery details.
5. Enable automatic updates and verify the operating system security patch level.
6. Disable installation from unknown sources and remove unneeded remote-access tools.
7. Configure transaction notifications and a low-risk transfer limit for the exercise account.
8. Record before-and-after settings without exposing secrets or recovery codes.

Evidence and acceptance

Hardening checklist with redacted screenshots and residual risks.

Acceptance: Credentials and recovery codes are absent; each control names the attack path it interrupts.

Activity 5 ScamShield Check and Report

Alignment: LO2 / A2 | Objective: Use ScamShield resources to check and report a simulated suspicious message safely.

Procedure

1. Open the official ScamShield site by typing the known government URL.
2. Review the current Check for Scams and reporting options.
3. Prepare the supplied simulated message, number and URL for checking.
4. Submit only the fictional training data supplied in the activity pack.
5. Record the result, confidence language and any follow-up advice.
6. Create a report using the simulated screenshot and redact unrelated personal data.
7. Compare the result with your own signal triage from Activity 1.
8. Document what ScamShield can support and what still requires independent judgment.

Evidence and acceptance

ScamShield check record, redacted report example and limitations note.

Acceptance: Only simulated data is submitted; the learner distinguishes a tool result from proof of legitimacy.

Activity 6 Financial Control Plan

Alignment: LO2 / A2 | Objective: Design preventive and detective controls for a personal or small-business payment workflow.

Procedure

1. Map the payment workflow from request to approval, authentication and settlement.
2. Identify where sender identity, account details and payment purpose are verified.
3. Add an independent callback for changed beneficiary or urgent payment requests.
4. Set a maker-checker rule and escalation threshold for high-risk transfers.
5. Choose notification, transfer-limit and Money Lock controls appropriate to the scenario.
6. Document exceptions and who may approve them.
7. Run the supplied fraudulent-invoice scenario through the revised workflow.
8. Record the blocked point, remaining exposure and recovery options.

Evidence and acceptance

Payment control map, approval matrix and scenario test result.

Acceptance: The design separates request, verification and approval, and names at least one preventive and one detective control.

Activity 7 Incident Evidence Pack

Alignment: LO3 / A3 | Objective: Preserve useful evidence and start containment without destroying the incident timeline.

Procedure

1. Start an incident log with date, time, timezone and reporter.
2. Preserve the original message, sender identifiers, URLs, screenshots and transaction references.
3. Record actions already taken, including clicks, downloads, credentials entered and transfers.
4. If compromise is suspected, isolate the affected device without factory-resetting it.
5. Use a separate trusted device to contact the bank and protect accounts.
6. Change exposed credentials in priority order and revoke active sessions.
7. List affected people, systems, data and financial accounts.
8. Package evidence with neutral filenames and a chain-of-custody note.

Evidence and acceptance

Incident timeline, evidence register and initial containment record.

Acceptance: The chronology is complete, evidence remains unaltered, and containment actions are distinguished from eradication.

Activity 8 Escalation Tabletop

Alignment: LO3 / A3 | Objective: Route a scam incident to the right internal and external parties using severity and evidence.

Procedure

1. Read the capstone scenario and assign an initial severity.
2. Identify immediate life-safety, financial, identity, device and business impacts.
3. Choose internal recipients: manager, security, finance, legal, data protection or communications.
4. Choose external recipients: bank, platform, ScamShield, Police or relevant dispute channel.
5. Prepare a concise escalation brief with verified facts, unknowns and requested action.
6. Sequence notifications to contain loss while preserving investigation options.
7. Update severity when the scenario inject introduces a second victim and public exposure.
8. Complete the after-action review with control improvements and owners.

Evidence and acceptance

Escalation matrix, incident brief, notification sequence and after-action actions.

Acceptance: The brief separates facts from assumptions and includes timestamps, evidence references, impact and a clear request.

Sources and Further Reading

[SPF Annual Scam and Cybercrime Brief 2025](#)

[SPF Mid-Year Scam and Cybercrime Brief 2026](#)

[ScamShield](#)

[OCC Online and Digital Scams](#)

[Fortinet Internet Fraud](#)

[Bitdefender Cyber Fraud](#)

[Europol Take Control of Your Digital Life](#)